Overview of Differential privacy machine learning

Subject: Differential privacy machine learning

1.

Attacks in practice Because differential privacy techniques are implemented on real computers, they are vulnerable to various attacks not possible to compensate for solely in the mathematics of the techniques themselves. In addition to standard defects of software artifacts that can be identified using testing or fuzzing, implementations of differentially private mechanisms may suffer from the following vulnerabilities:

2.

The 2006 Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam D. Smith article introduced the concept of ε-differential privacy, a mathematical definition for the privacy loss associated with any data release drawn from a statistical database. (Here, the term statistical database means a set of data that are collected under the pledge of confidentiality for the purpose of producing statistics that, by their production, do not compromise the privacy of those individuals who provided the data.) The definition of ε-differential privacy requires that a change to one entry in a database only creates a small change in the probability distribution of the outputs of measurements, as seen by the attacker. The intuition for the definition of ε-differential privacy is that a person's privacy cannot be compromised by a statistical release if their data are not in the database. In differential privacy, each individual is given roughly the same privacy that would result from having their data removed. That is, the statistical functions run on the database should not be substantially affected by the removal, addition, or change of any individual in the data. How much any individual contributes to the result of a database query depends in part on how many people's data are involved in the query. If the database contains data from a single person, that person's data contributes 100%. If the database contains data from a hundred people, each person's data contributes just 1%. The key insight of differential privacy is that as the query is made on the data of fewer and fewer people, more noise needs to be added to the query result to produce the same amount of privacy. Hence the name of the 2006 paper, "Calibrating Noise to Sensitivity in Private Data Analysis."

3.

Definition Let ε be a positive real number and A {\displaystyle {\mathcal {A}}} be a randomized algorithm that takes a dataset as input (representing the actions of the trusted party holding the data). Let im A {\displaystyle {\textrm {im}}\ {\mathcal {A}}} denote the image of A {\displaystyle {\mathcal {A}}} . The algorithm A {\displaystyle {\mathcal {A}}} is said to provide (ε, δ)-differential privacy if, for all datasets D 1 {\displaystyle D_{1}} and D 2 {\displaystyle D_{2}} that differ on a single element (i.e., the data of one person), and all subsets S {\displaystyle S} of im A {\displaystyle {\textrm {im}}\ {\mathcal {A}}} :

4.

which is at most e | f (D 1) − f (D 2) | λ ≤ e Δ (f) λ {\displaystyle e^{\frac {|f(D_{1})-f(D_{2})|}{\lambda }}\leq e^{\frac {\Delta (f)}{\lambda }}\,\!} . We can consider Δ (f) λ {\displaystyle {\frac {\Delta (f)}{\lambda }}\,\!} to be the privacy factor ε {\displaystyle \varepsilon

\,\!} . Thus T {\displaystyle {\mathcal {T}}\,\!} follows a differentially private mechanism (as can be seen from the definition above). If we try to use this concept in our diabetes example then it follows from the above derived fact that in order to have A {\displaystyle {\mathcal {A}}\,\!} as the ε {\displaystyle \varepsilon \,\!} -differential private algorithm we need to have λ = 1 / ε {\displaystyle \lambda =1/\varepsilon \,\!} . Though we have used Laplace noise here, other forms of noise, such as the Gaussian Noise, can be employed, but they may require a slight relaxation of the definition of differential privacy.

5.

ε-differentially private mechanisms Since differential privacy is a probabilistic concept, any differentially private mechanism is necessarily randomized. Some of these, like the Laplace mechanism, described below, rely on adding controlled noise to the function that we want to compute. Others, like the exponential mechanism and posterior sampling sample from a problem-dependent family of distributions instead. An important definition with respect to ε-differentially private mechanisms is sensitivity. Let d {\displaystyle d} be a positive integer, D {\displaystyle {\mathcal {D}}} be a collection of datasets, and f : D → R d {\displaystyle f\colon {\mathcal {D}}\rightarrow \mathbb {R} ^{d}} be a function. One definition of the sensitivity of a function, denoted Δ f {\displaystyle \Delta f} , can be defined by: Δ f = max ■ f (D 1) − f (D 2) ■ 1 , {\displaystyle \Delta f=\max \lVert f(D_{1})-f(D_{2})\rVert _{1},} where the maximum is over all pairs of datasets D 1 {\displaystyle D_{1}} and D 2 {\displaystyle D_{2}} in D {\displaystyle {\mathcal {D}}} differing in at most one element and ■ · ■ 1 {\displaystyle \lVert \cdot \rVert _{1}} denotes the L1 norm. In the example of the medical database below, if we consider f {\displaystyle f} to be the function Q i {\displaystyle Q_{i}} , then the sensitivity of the function is one, since changing any one of the entries in the database causes the output of the function to change by either zero or one. This can be generalized to other metric spaces (measures of distance), and must be to make certain differentially private algorithms work, including adding noise from the Gaussian distribution (which requires the L2 norm) instead of the Laplace distribution. There are techniques (which are described below) using which we can create a differentially private algorithm for functions, with parameters that vary depending on their sensitivity.

6.

A {\displaystyle A} and B {\displaystyle B} (where A {\displaystyle A} contains the group and B {\displaystyle B} does not). In this case M ■ T {\displaystyle M\circ T} is (ε × c × h) {\displaystyle (\varepsilon \times c\times h)} -differentially private.

7.

Stable transformations A transformation T {\displaystyle T} is c {\displaystyle c} -stable if the Hamming distance between T (A) {\displaystyle T(A)} and T (B) {\displaystyle T(B)} is at most c {\displaystyle c} -times the Hamming distance between A {\displaystyle A} and B {\displaystyle B} for any two databases A , B {\displaystyle A,B} . If there is a mechanism M {\displaystyle M} that is ε {\displaystyle \varepsilon } -differentially private, then the composite mechanism M ■ T {\displaystyle M\circ T} is (ε × c) {\displaystyle (\varepsilon \times c)} -differentially private. This could be generalized to group privacy, as the group size could be thought of as the Hamming distance h {\displaystyle h} between